Mortality, Growth, and the Existential-Risk Cutoff

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Abstract

We characterise the maximum extinction hazard δ^* at which a transformative-technology scenario delivers higher social welfare than a status-quo baseline. The framework is a standard CRRA representative-agent economy with three primitives: a growth rate g, an individual mortality rate m, and a per-period flow utility calibrated against empirical Value of Statistical Life (VSL) data. We solve for δ^* by numerical root-finding on the welfare-equality condition $W_{AI}(\delta^*) = W_0$. The main finding: under the standard calibration ($\gamma = 2$, $\rho = \rho_s = 1\%$, $v_{c_0} = 6$), the welfare gain that determines the cutoff comes overwhelmingly from the mortality channel (approximately 94%), with the consumption-growth channel contributing the remaining 6%. At the headline parameters ($g_{AI} = 10\%$, $m_{AI} = 0.5\%$, mortality halved from baseline), $\delta^* \approx 0.20\%$ per year. Without mortality improvement ($m_{AI} = m_0$), the cutoff collapses to approximately 0.015% at $\gamma = 2$: growth alone does not buy meaningful extinction-hazard tolerance. The cutoff is highly sensitive to the social discount rate (δ^* falls by approximately the same factor as ρ_s) and weakly sensitive to the VSL anchor v_{c_0} (the anchor scales both W_0 and W_{AI}, leaving the ratio approximately invariant). The framework is a structured cost-benefit calculation, not a recommendation about any specific technology. Its policy use is to clarify which empirical disagreements (mortality assumptions, discount-rate choice) drive the cutoff and which (growth-rate assumptions) do not.

Keywords: existential risk, value of statistical life, social discounting, mortality reduction, transformative AI
JEL: D81, H43, O40, Q54

1. Introduction

The economics of existential risk asks how to value the avoidance of low-probability, high-magnitude extinction outcomes within a welfare-economic framework. Aschenbrenner (2020) and Trammell (2020) provide the foundational analyses; Jones (2024) extends the framework to growth-risk complementarities; the broader cost-benefit-of-catastrophic-risk tradition is summarised by Posner (2004) and Nordhaus (2011). The central methodological question is how the various channels (growth gains, mortality reduction, extinction probability, social discounting) combine to determine a policy-relevant threshold.

This paper provides a structured cost-benefit calculation of the threshold and asks a sharper question of it: *which channel does the work*? Under what circumstances is the cutoff δ^* at which a transformative-technology scenario equals the status-quo baseline in welfare driven by faster growth, and under what circumstances is it driven by mortality reduction, or by social discounting choices? The structural decomposition delivers a clean answer that is at odds with the intuitive "growth versus extinction" framing.

The model is a standard representative-agent CRRA economy with individual mortality rate m, growth rate g, social discount rate ρ_s, and a per-period flow utility $\bar{u}$ calibrated to empirical VSL data. We compare a baseline at $(g_0, m_0) = (2\%, 1\%)$ to a transformative-technology scenario at (g_{AI}, m_{AI}). The extinction hazard δ enters as an additional Poisson rate that augments both the individual mortality rate (each cohort discounts by $\rho + m + \delta$) and the social discount rate (future cohorts discount by $\rho_s + \delta$). The cutoff δ^* is the value at which $W_{AI}(\delta^*) = W_0$, solved by numerical root-finding.

Main results. (i) The mortality channel dominates. At the standard calibration ($\gamma = 2$, $\rho = \rho_s = 1\%$, $v_{c_0} = 6$, $g_{AI} = 10\%$, $m_{AI} = 0.5\%$), the welfare gain $W_{AI}(0) - W_0$ decomposes as 94% from the flow-utility (mortality-driven) channel and 6% from the consumption (growth-driven) channel. The cutoff $\delta^* \approx 0.20\%$ per year reflects this decomposition: it is overwhelmingly a calculation of life-years preserved per unit hazard, not of consumption gained per unit hazard.

(ii) Growth alone produces a near-zero cutoff. If the transformative scenario delivers no mortality improvement ($m_{AI} = m_0 = 1\%$), the cutoff $\delta^* \approx 0.015\%$ at $\gamma = 2$ and is essentially zero at $\gamma = 3$. Doubling g_{AI} from 10% to

20% raises δ^* by less than 10%. Growth, in this framework, does not buy meaningful extinction-hazard tolerance.

(iii) The cutoff is structural in ρ_s, not in v_{c_0}. A tenfold reduction in ρ_s (from 1% to 0.1%) reduces δ^* by approximately the same factor: the cutoff scales with ρ_s at leading order. By contrast, varying v_{c_0} from 2 to 10 (a fivefold range covering all empirically defensible VSL calibrations) changes δ^* by less than 15%: the VSL anchor enters W_0 and W_{AI} approximately symmetrically and largely cancels in the ratio.

Methodological positioning. The framework is a structured cost-benefit calculation. The output is a parametric cutoff, not a recommendation. Three things the framework does well: (a) it makes the structural sensitivities transparent, including the dominant role of the mortality channel that informal "growth versus extinction" framings obscure; (b) it identifies which empirical disagreements (about ρ_s, m_{AI}) translate to large cutoff differences and which (about v_{c_0}, g_{AI}) do not; (c) it provides a benchmark against which empirical estimates of δ can be assessed. Three things the framework does not do: (a) estimate the actual extinction hazard from any specific technology; (b) account for learning about δ over time, which under irreversibility would lower the effective adoption threshold below the static δ^* here; (c) account for growth-and-hazard complementarities (Jones, 2024).

Plan. Section 2 positions the paper. Section 3 sets out the model and the VSL calibration. Section 4 derives the cutoff and the decomposition that delivers the main finding. Section 5 presents the quantitative results. Section 6 discusses what the framework can and cannot say. Section 7 concludes.

2. Related literature

Existential-risk economics. Aschenbrenner (2020) and Trammell (2020) embed extinction hazards in endogenous-growth frameworks. Jones (2024) introduces growth-risk complementarity (the technology that drives growth may itself raise hazard). The present paper is closer to the static / partial-equilibrium cost-benefit tradition and abstracts from the complementarity; we return to this caveat in Section 6.

Value of statistical life. Viscusi (2018) and Viscusi and Aldy (2003) summarise the empirical VSL literature. UK and US estimates anchor the calibration:

VSL $\approx$ £8.59m and \$10m respectively, normalised by remaining life expectancy and per-capita consumption to give $v_{c_0} \in [6, 8.3]$. The additive flow benefit $\bar{u}$ through which VSL enters the utility function is the value-of-a-life-year object studied by Rosen (1988), Murphy and Topel (2006), and Hall and Jones (2007); the last, in particular, derives an additive value-of-life term from standard preferences and is the closest antecedent to our $\bar{u}$.

Social discounting. The debate over ρ_s has its origin in Ramsey (1928) and runs from Stern (2007) (near-zero ρ_s on intergenerational-equity grounds) through Nordhaus (2007, 2017) (positive ρ_s calibrated to market rates) to Weitzman (1998, 2007) (parameter uncertainty implies effective discounting at the lowest-plausible rate). We do not take a position on ρ_s; we present results across a range and identify the cutoff's dependence on it as a key structural sensitivity.

Catastrophic-risk policy. Posner (2004) and Nordhaus (2011) frame the broader policy-design question, and Weitzman (2009) and Martin and Pindyck (2015) analyse the cost-benefit treatment of low-probability, high-magnitude catastrophes of the kind studied here. The present paper sits within this tradition but focuses narrowly on the structural decomposition of δ^*.

3. Model

What carries the result. The conclusion rides on a small set of world inputs: the VSL anchor (which calibrates the value-of-life constant $\bar{u}$ below), the scenario parameters (growth g, mortality m, and the extinction hazard δ), and the social discount rate ρ_s. The functional choices around them are structural, not substantive: the CRRA-plus-additive-flow-benefit form (the additive $\bar{u}$ is the device that lets VSL enter utility, not a claim about its shape), the cohort-integral social-welfare aggregation, and the closed-form discount factors D_1–D_4. The mortality-channel dominance of theorem 2 is accordingly a statement about those world inputs — the size of $\bar{u}$ relative to consumption utility and the smallness of ρ_s — not an artefact of the functional form.

3.1. Preferences and value of life

Flow utility takes the standard CRRA form with a per-period flow benefit:

$$v(c) \;=\; \frac{c^{1-\gamma}}{1-\gamma} + \bar{u}, \qquad \gamma > 0, \gamma \neq 1, \tag{1}$$

with the logarithmic limit $v(c) = \log c + \bar{u}$ at $\gamma = 1$. The constant $\bar{u} > 0$ is a flow benefit independent of consumption: the value of being alive that is not reducible to consumption flow.

VSL calibration of $\bar{u}$. Empirically, VSL estimates the dollar value of a marginal mortality risk reduction. The per-life-year value normalised by per-capita consumption is

$$v_{c_0} \;\equiv\; \frac{\text{VSL}}{\text{Life Expectancy} \times \text{Consumption per capita}}. \tag{2}$$

For the UK (VSL $\approx$ £8.59m, $\approx$ 41 remaining years, $\approx$ £25,373 per capita): $v_{c_0} \approx 8.26$. For the US (\$10m, $\approx$ 40 years, \$41,667): $v_{c_0} \approx 6.00$. We adopt $v_{c_0} = 6$ as the baseline with sensitivity reported.

Normalising $c_0 = 1$, the constant $\bar{u}$ is pinned by $v(c_0) = v_{c_0}$:

$$\bar{u} \;=\; v_{c_0} - \frac{1}{1-\gamma}, \tag{3}$$

giving $\bar{u} = 7$ at $\gamma = 2$ and $\bar{u} = 6.5$ at $\gamma = 3$. The logarithmic limit gives $\bar{u} = v_{c_0} = 6$ at $\gamma = 1$.

3.2. Individual lifetime utility

The individual faces a per-period mortality hazard m and discounts the future at rate ρ. Consumption grows at g. Expected lifetime utility starting at c_0 is

$$U(g, m) \;=\; \int_0^\infty e^{-(\rho+m)t}\, v(c_0 e^{gt})\, dt. \tag{4}$$

For CRRA utility with constant growth, this evaluates in closed form to

$$U(g, m) \;=\; \frac{c_0^{1-\gamma}}{(1-\gamma)\big[(\rho+m) + (\gamma-1)g\big]} \;+\; \frac{\bar{u}}{\rho+m}, \tag{5}$$

under the convergence condition $(\rho + m) + (\gamma - 1)g > 0$ (an upper bound on g when $\gamma > 1$). The first term is the consumption-utility integral; the second is the constant-flow-benefit integral.

3.3. *Social welfare*

The social planner discounts future generations at rate ρ_s and integrates lifetime utility over future cohorts, each starting at consumption $c_0 e^{gt}$:

$$W(g,m) \;=\; \int_0^\infty e^{-\rho_s t}\, U_t(g,m)\, dt \;=\; \frac{c_0^{1-\gamma}}{(1-\gamma)\, D_1(g,m)\, D_2(g)} \;+\; \frac{\bar{u}}{D_3(m)\, D_4}, \tag{6}$$

where we define the four discount factors:

$$\begin{aligned} D_1(g,m) &\equiv (\rho+m)+(\gamma-1)g, & D_2(g) &\equiv \rho_s+(\gamma-1)g, \\ D_3(m) &\equiv \rho+m, & D_4 &\equiv \rho_s. \end{aligned} \tag{7}$$

D_1 is the cohort discount rate for consumption-utility; D_2 is the social discount rate adjusted for cross-cohort consumption growth; D_3 and D_4 are the analogous factors for the constant flow-utility term.

Extinction hazard. An extinction-hazard rate δ adds a Poisson-arrival risk of civilisation ending. The hazard enters W in two places: it augments individual mortality (each cohort discounts at $\rho+m+\delta$ in D_1, D_3) and it augments social discounting (future cohorts that do not exist contribute zero, so $\rho_s+\delta$ in D_2, D_4):

$$W_{AI}(\delta) \;=\; \frac{c_0^{1-\gamma}}{(1-\gamma)\, D_1(g_{AI}, m_{AI}+\delta)\, D_2'(g_{AI},\delta)} \;+\; \frac{\bar{u}}{D_3(m_{AI}+\delta)\,(\rho_s+\delta)}, \tag{8}$$

where $D_2'(g,\delta) \equiv (\rho_s+\delta)+(\gamma-1)g$.

4. The existential-risk cutoff and its decomposition

Definition 1 (Existential-risk cutoff)**.** The cutoff δ^* is the value of δ at which $W_{AI}(\delta^*) = W_0$, where $W_0 \equiv W(g_0, m_0)$. For $\delta < \delta^*$, $W_{AI}(\delta) > W_0$ (transformative scenario welfare-superior); for $\delta > \delta^*$, $W_{AI}(\delta) < W_0$.

The cutoff has no useful closed form (it is the root of a rational function of δ) and is computed by numerical root-finding.

4.1. *Welfare-gain decomposition*

The substantive structural finding emerges from decomposing the welfare gain $W_{AI}(0) - W_0$ (transformative scenario at zero hazard versus baseline) into the two channels of (6):

$$W_{AI}(0) - W_0 \;=\; \underbrace{\frac{c_0^{1-\gamma}}{1-\gamma}\left[\frac{1}{D_1^{AI}D_2^{AI}} - \frac{1}{D_1^0 D_2^0}\right]}_{\text{consumption (growth) channel}} \;+\; \underbrace{\bar{u}\left[\frac{1}{D_3^{AI}D_4} - \frac{1}{D_3^0 D_4}\right]}_{\text{flow-utility (mortality) channel}}, \tag{9}$$

where the D^{AI} are evaluated at (g_{AI}, m_{AI}) and the D^0 at (g_0, m_0).

Why the channels are asymmetric. The consumption channel scales with $c_0^{1-\gamma}/(1-\gamma)$, which equals -1 at $\gamma = 2$ with $c_0 = 1$. The flow-utility channel scales with $\bar{u} = 7$ at $\gamma = 2$. Both channels also scale with the difference between D^{AI} and D^0 factors. The consumption-channel denominators include $(\gamma - 1)g \in [0.02, 0.10]$ (substantial), so the percentage change in $D_1 D_2$ between scenarios is modest. The flow-utility-channel denominators are $D_3 D_4 = (\rho + m)\rho_s$, with $\rho_s = 0.01$ tiny, so even small changes in m produce large percentage changes in $1/(D_3 D_4)$.

The combination: $\bar{u}$ is larger than $|c_0^{1-\gamma}/(1-\gamma)|$ by a factor of approximately 7; and the percentage swing in the mortality denominators is larger than in the consumption denominators by a factor of approximately 2–3 because ρ_s enters the flow-utility denominator as a multiplier and is small. The flow-utility channel therefore dominates by roughly an order of magnitude.

Proposition 2 (Mortality channel dominates the welfare-gain decomposition)**.** *At the standard calibration ($\gamma = 2$, $\rho = \rho_s = 1\%$, $v_{c_0} = 6$, $g_0 = 2\%$, $m_0 = 1\%$, $g_{AI} = 10\%$, $m_{AI} = 0.5\%$), the flow-utility channel contributes* 93.9% *of $W_{AI}(0) - W_0$; the consumption channel contributes* 6.1%.

Proof. Direct computation from (9) with the given parameters. Consumption-channel value: 754.28. Flow-utility-channel value: 11,666.67. Ratio: $11667/(11667 + 754) = 0.939$. Verified by direct computation. □

Implication for the cutoff. The cutoff δ^* inherits the channel dominance. Because the welfare gain that δ^* trades against is overwhelmingly flow-utility-driven, the cutoff is essentially the answer to "how much extinction-hazard can the mortality-improvement-driven welfare gain absorb before the comparison flips?" Faster growth contributes a small additional welfare gain that allows a slightly higher cutoff, but the structural answer is determined by mortality, not by growth.

This is the substantive finding of the paper and it is at odds with the intuitive "growth versus extinction" framing.

5. Quantitative results

5.1. *The headline table: δ^* across scenarios*

Table 1 reports δ^* across a grid of (g_{AI}, m_{AI}, γ) at the baseline calibration ($\rho = \rho_s = 1\%$, $v_{c_0} = 6$, $g_0 = 2\%$, $m_0 = 1\%$).

g_{AI}	m_{AI}	γ	δ^* (%/yr)	Flow-channel share	u_{bar}
5%	1.0%	1	0.228	0.00	6.0
5%	1.0%	2	0.012	0.00	7.0
5%	1.0%	3	0.003	0.00	6.5
5%	0.5%	1	0.412	0.37	6.0
5%	0.5%	2	0.198	0.95	7.0
5%	0.5%	3	0.189	0.99	6.5
10%	1.0%	1	0.491	0.00	6.0
10%	1.0%	2	0.015	0.00	7.0
10%	1.0%	3	0.003	0.00	6.5
10%	0.5%	1	0.683	0.20	6.0
10%	0.5%	2	0.201	0.94	7.0
10%	0.5%	3	0.189	0.99	6.5
20%	1.0%	1	0.853	0.00	6.0
20%	1.0%	2	0.016	0.00	7.0
20%	1.0%	3	0.003	0.00	6.5
20%	0.5%	1	1.053	0.11	6.0
20%	0.5%	2	0.203	0.93	7.0
20%	0.5%	3	0.190	0.99	6.5

Table 1. Existential-risk cutoffs δ^* (% per year) across scenarios. "Flow-channel share" is the share of $W_{AI}(0) - W_0$ contributed by the flow-utility (mortality) channel. At $\gamma \geq 2$ with mortality improvement, this share is ≥ 0.94: mortality dominates the cutoff. Without mortality improvement, the cutoff at $\gamma \geq 2$ falls to near zero regardless of growth.

Reading the table. (i) Without mortality improvement ($m_{AI} = 1\%$), the cutoff at $\gamma \geq 2$ is essentially zero. At $\gamma = 2$, doubling growth from 10% to 20% raises δ^* from 0.015% to 0.016%. Growth alone does not buy meaningful hazard tolerance.

(ii) With mortality halved ($m_{AI} = 0.5\%$), the cutoff jumps an order of magnitude. At $\gamma = 2$, $g_{AI} = 10\%$: δ^* rises from 0.015% (no mortality improvement) to 0.20% (mortality halved). The cutoff is overwhelmingly about life-years preserved.

(iii) The $\gamma = 1$ row is qualitatively different from $\gamma \geq 2$. At $\gamma = 1$ (log utility), the consumption-channel share is comparable to the flow-utility-channel share, and growth does substantial work in raising δ^*. This reflects the structural difference between log utility and CRRA with $\gamma > 1$: log utility has unbounded marginal utility at low c and a slower-falling marginal utility at high c, making consumption growth contribute more to welfare than under $\gamma > 1$. The standard macro calibration is $\gamma = 2$; the log-utility case is reported for completeness but does not match standard practice.

5.2. Sensitivity to social discounting

Table 2 reports δ^* across values of ρ_s at the headline scenario ($g_{AI} = 10\%$, $m_{AI} = 0.5\%$, $\gamma = 2$, $v_{c_0} = 6$).

ρ_s	δ^* (%/yr)	Flow-channel share
0.05% (Stern)	0.016	0.995
0.10%	0.031	0.991
0.50%	0.124	0.962
1.00% (standard)	0.201	0.939
3.00%	0.357	0.900

Table 2. Cutoff δ^* as a function of social discount rate ρ_s. The cutoff falls approximately proportionally with ρ_s: lower social discounting raises the weight on distant future generations and makes the same hazard more costly per period. The flow-channel share rises toward 1 as ρ_s falls because the flow-utility denominator ρ_s becomes smaller relative to the consumption-channel denominator $\rho_s + (\gamma - 1)g$.

The structural reason for the sensitivity. The flow-utility denominator is $D_4 = \rho_s$ (no growth adjustment). At $\rho_s = 0.05\%$, $1/D_4 = 2000$; at $\rho_s = 1\%$, $1/D_4 = 100$. The flow-utility channel scales as $\bar{u}/D_4$ in absolute terms, so reducing ρ_s by a factor of 20 raises the flow-utility-channel magnitude by the same factor. The welfare gain $W_{AI}(0) - W_0$ scales similarly. But so does the rate at which $W_{AI}(\delta)$ falls with δ (because δ enters D_4). The cutoff δ^* is the ratio of these two quantities, which scales approximately with ρ_s. Hence the cutoff inherits the ρ_s-scaling.

5.3. Sensitivity to the VSL anchor

Table 3 reports δ^* across values of v_{c_0} at the headline scenario.

v_{c_0}	$\bar{u}$	δ^* (%/yr)	Flow-channel share
2	3	0.223	0.87
4	5	0.208	0.92
6	7	0.201	0.94
8	9	0.198	0.95
10	11	0.196	0.96

Table 3. Cutoff δ^* as a function of VSL anchor v_{c_0} at the headline scenario. The cutoff is weakly sensitive to v_{c_0}: a fivefold variation ($v_{c_0} \in [2, 10]$, covering all empirically defensible VSL ranges) changes δ^* by less than 14%. The structural reason: v_{c_0} scales $\bar{u}$, which scales both W_0 and W_{AI} similarly in absolute terms; the cutoff δ^* which is set by the ratio is approximately invariant.

Asymmetry of the structural sensitivities. The cutoff is structural in ρ_s and in m_{AI} (the difference $m_0 - m_{AI}$ that drives the flow-utility-channel welfare gain), but not in v_{c_0} or in g_{AI}. A policy question that turns on ρ_s or on m_{AI} is one the framework can sharpen. A policy question that turns on v_{c_0} or on g_{AI} is one the framework cannot sharpen because the cutoff is weakly responsive to those parameters at the relevant ranges.

6. Discussion

The framework's central finding. At standard macro calibrations ($\gamma \geq 2$), the existential-risk cutoff is overwhelmingly determined by the mortality channel. Growth contributes secondarily; without mortality improvement, growth alone does not buy meaningful extinction-hazard tolerance. The informal "growth versus extinction" framing common in popular discussions of transformative-AI policy is misleading: the relevant trade-off is "mortality reduction versus extinction," with growth as a minor modifier.

This is a substantive finding about what the model says, not a recommendation. Two readings are possible.

(i) The model is right and the popular framing is wrong. If you think CRRA with $\gamma = 2$ is the right way to model preferences and $v_{c_0} = 6$ is the right VSL calibration, then the policy debate should be reframed around mortality and discount-rate choices, not around growth.

(ii) The model is wrong and the popular framing is reasonable. If you think growth should matter more, you need a model in which it does. Three structural changes would each have this effect: (a) a non-CRRA utility specification that gives more weight to consumption growth at high c (e.g., habits, Stone-Geary, narrowly-bounded utility); (b) explicit modelling of the welfare-relevant outputs of growth beyond consumption (e.g., spillovers to health and longevity, which would re-route growth back through the flow-utility channel); (c) a complementarity between growth and mortality reduction, in which case the two channels cannot be separated in the way (9) does.

The first and third are plausible; the second is empirically defensible (better technology has historically reduced mortality, and the longer-run welfare gain from growth runs partly through that). Each would alter the headline finding. The present framework provides a benchmark against which these extensions can be compared: the question is how much the headline 94% flow-channel share would shift under each.

Caveats common to all cutoff calculations. Parameter uncertainty. Under Weitzman (2007)-style uncertainty, the effective social discount rate is the lowest plausible value, not the mean. The headline cutoff under standard ρ_s is therefore an upper bound on the policy-relevant threshold under uncertainty.

Learning and irreversibility. The framework treats δ as known. Under uncertainty about δ and irreversibility of extinction, the option value of delaying adoption to gather information is positive. The framework does not capture this; a real-options extension would lower the effective adoption threshold below the static δ^*.

Growth-and-hazard complementarity. Jones (2024) argues that the technological capability that delivers growth may itself raise the hazard. Under this complementarity, g_{AI} and δ are not independent and the cutoff calculation is not separable in the way (9) suggests. The strength of the complementarity is empirically unknown.

Methodological positioning. The framework belongs to the structured-cost-benefit tradition. It does not estimate the actual extinction hazard from any specific technology. Its value is in clarifying which empirical disagreements drive the cutoff (mortality assumptions, social discounting) and which do not (VSL anchor, growth rate at standard parameters). Reasonable policymakers and analysts who disagree about extinction hazard can use the framework to identify which underlying parameter disagreement is driving the policy disagreement.

7. Conclusion

A standard CRRA representative-agent framework augmented by mortality and VSL calibration delivers a structural existential-risk cutoff δ^*. We solve for it by numerical root-finding on the welfare-equality condition $W_{AI}(\delta^*) = W_0$.

The headline finding is the decomposition: at standard calibrations, the mortality channel contributes approximately 94% of the welfare gain that the cutoff trades against extinction-hazard, with the consumption-growth channel contributing the remaining 6%. The cutoff is therefore essentially a calculation of life-years preserved per unit hazard, not of consumption gained per unit hazard. Without mortality improvement, growth alone produces a near-zero cutoff at $\gamma \geq 2$. The cutoff is structural in the social discount rate (scaling proportionally with ρ_s) but weakly sensitive to the VSL anchor (which scales both sides of the comparison approximately symmetrically).

The framework does not estimate the actual extinction hazard from any specific technology. It provides a structured benchmark in which the relative importance of mortality, growth, discounting, and VSL choices can be made transparent. The policy implication is that the relevant debate is not "growth versus extinction" but "mortality reduction and social discounting versus extinction," with the choice of ρ_s identified as the dominant structural sensitivity.

References

Aschenbrenner, L., 2020. Existential risk and growth. Working paper. Global Priorities Institute, Oxford.

Hall, R.E., Jones, C.I., 2007. The value of life and the rise in health spending. Quarterly Journal of Economics 122, 39–72.

Jones, C.I., 2024. The a.i. dilemma: Growth versus existential risk. American Economic Review: Insights 6, 575–590.

Martin, I.W.R., Pindyck, R.S., 2015. Averting catastrophes: The strange economics of scylla and charybdis. American Economic Review 105, 2947–2985.

Murphy, K.M., Topel, R.H., 2006. The value of health and longevity. Journal of Political Economy 114, 871–904.

Nordhaus, W.D., 2007. A review of the stern review on the economics of climate change. Journal of Economic Literature 45, 686–702.

Nordhaus, W.D., 2011. The economics of tail events with an application to climate change. Review of Environmental Economics and Policy 5, 240–257.

Nordhaus, W.D., 2017. Revisiting the social cost of carbon. Proceedings of the National Academy of Sciences 114, 1518–1523.

Posner, R.A., 2004. Catastrophe: Risk and Response. Oxford University Press.

Ramsey, F.P., 1928. A mathematical theory of saving. Economic Journal 38, 543–559.

Rosen, S., 1988. The value of changes in life expectancy. Journal of Risk and Uncertainty 1, 285–304.

Stern, N., 2007. The Economics of Climate Change: The Stern Review. Cambridge University Press.

Trammell, P., 2020. Existential risk and exogenous growth. Working paper.

Viscusi, W.K., 2018. Pricing Lives: Guideposts for a Safer Society. Princeton University Press.

Viscusi, W.K., Aldy, J.E., 2003. The value of a statistical life: a critical review of market estimates throughout the world. Journal of Risk and Uncertainty 27, 5–76.

Weitzman, M.L., 1998. Why the far-distant future should be discounted at its lowest possible rate. Journal of Environmental Economics and Management 36, 201–208.

Weitzman, M.L., 2007. A review of the stern review on the economics of climate change. Journal of Economic Literature 45, 703–724.

Weitzman, M.L., 2009. On modeling and interpreting the economics of catastrophic climate change. Review of Economics and Statistics 91, 1–19.